

Software Program Plan - Revision 3

Date: 2026-03-13
Project: Agent Follow-Up

1. Purpose

Deliver a provable follow-up system for OpenClaw background work that removes reliance on agent memory and replaces it with a scheduler-backed, evidence-producing mechanism.

2. Implementation Readiness Goal

Revision 3 is intended to close the remaining gate items from the Rev 2 review so that Phase 1 implementation can begin without unresolved interface ambiguity.

3. Phase 1 Decisions

Area	Chosen Direction
Scheduler	OpenClaw native cron (openclaw cron)
Emitter	<code>openclaw message send --channel telegram --target <id> --message "..."</code>
Registry	SQLite-backed local state store
Evaluation	Deterministic, non-LLM watchdog
Phase 1 task support	Process-backed and marker-backed tasks only

4. Program Risks

ID	Risk	Mitigation
R-1	OpenClaw delegated-session status is not queryable enough for Phase 1	Defer delegated-session observation to Phase 2 and require explicit completion markers before supporting it.
R-2	Notification duplication or spam	Use explicit notification timestamps/status plus deterministic suppression logic.
R-3	Scheduler behavior differs from assumptions	Use OpenClaw native cron and verify with job/run evidence before relying on it.
R-4	Message delivery failures	Record emitter result and surface degraded-state evidence rather than claiming delivery.
R-5	Marker files not written by the observed task	Phase 1 registration contract requires owned marker paths for process-backed tasks.

5. Deliverables

- Node.js CLI for registration/update/resolve helpers
- SQLite schema and migration bootstrap
- OpenClaw cron watchdog job definition
- Notification emitter wrapper
- Verification proof package

6. Success Criteria

- Tracked tasks produce proactive updates without operator prompting.

- At least one completion, one overdue reminder, and one failure path are proven.
- The system survives restart with durable task state intact.
- The system uses OpenClaw-native cron and OpenClaw-native messaging in Phase 1.